

Xalan XSLT 3.0 specification implementation status

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(1) XSL Transformations (XSLT) 3.0 and XML Path Language (XPath) 3.1

The XSLT 3.0 specification defines the following conformance features, and the level upto which Xalan implements them.

- | | |
|-----------------------------------|--|
| a) Basic XSLT processor | Supported |
| | XSLT 3.0 instructions and XPath language features, whose implementations are available are described in subsequent sections of this document, below. |
| b) Schema aware XSLT processor | Supported |
| | An XML Schema document can be imported into an XSL stylesheet using <code>xsl:import-schema</code> instruction, and schema's global type definitions and element & attribute declarations can be used within the stylesheet. |
| | Schema aware feature where XML input document, resulting in node tree having detailed type annotations on all possible nodes is not supported. i.e, XPath processor is natively not schema aware. |
| c) Serialization feature | Supported |
| | A new support for <code>xsl:output method="json"</code> is available, in addition to existing <code>xsl:output method</code> values. |
| d) Streaming feature | Not supported |
| e) Dynamic evaluation feature | Supported |
| f) XPath 3.1 feature, for arrays | Supported |
| g) Higher-order functions feature | Supported |

Following are details of XSL 3.0 family of language features, whose working implementation is available on Xalan's dev repos branch 'xalan-j_xslt3.0_mvn':

1.1) XSLT 3.0

XSLT version 3.0 specification : <https://www.w3.org/TR/xslt-30/>

- 1) `xsl:for-each-group` instruction
- 2) `xsl:analyze-string` instruction
- 3) `xsl:iterate` instruction
- 4) `xsl:for-each` instruction implementation improvements, for new XSLT 3.0 requirements. Particularly, `xsl:for-each` instruction being able to iterate XPath atomic values in addition to nodes.
- 5) `xsl:evaluate` instruction
- 6) `xsl:function` instruction
- 7) `xsl:sequence` instruction
- 8) The following XSL stylesheet elements can now have attributes ‘type’ and ‘validation’ : `xsl:element`, literal result element (`xsl:validation` and `xsl:attribute`), `xsl:attribute`, `xsl:copy-of`, `xsl:copy`.
- 9) `xsl:attribute` element can now have both, "select" attribute and child sequence constructor. But only one of these is allowed to be present on `xsl:attribute` instruction as specified by XSLT 3.0 specification.
- 10) `xsl:import-schema` instruction
- 11) `xsl:variable` instruction evaluation to node set instead of result tree fragment (RTF). This XSLT specification change was first introduced in XSLT 2.0. With XSLT 1.0, if RTF has to be used as node set, then it has to be converted to node set using node-set extension function.
- 12) The sequence type expression "as" attribute on XSLT elements `xsl:variable`, `xsl:template`, `xs:function`, `xsl:param`, `xsl:with-param`, `xsl:evaluate`.
- 13) XSL template tunnel parameters
- 14) `xsl:value-of` instruction can now produce result either via its “select” attribute, or by `xsl:value-of` instruction’s child sequence constructor. `xsl:value-of` instruction can now have an attribute named ‘separator’ as well.
- 15) `xsl:merge` instruction
- 16) `xsl:fork` instruction
- 17) `xsl:source-document` instruction
- 18) `xsl:try` and `xsl:catch` instructions

- 19) `xsl:character-map` instruction
- 20) Specifying XSL transformation's initial template name and support for initial template's default name `xsl:initial-template`.
- 21) XSLT function implementations
 - a) New function implementations : `fn:current-grouping-key`, `fn:current-group`, `fn:regex-group`, `fn:current-merge-group`, `fn:current-merge-key`
 - b) Function implementation enhancements : `fn:system-property`
- 22) Implementation of XSLT 3.0 attributes `xpath-default-namespace` and `expand-text`
- 23) `xsl:mode` instruction

Support for following new Xalan XSL transformation properties:

`http://apache.org/xalan/validation` (used to enable XML input document validation when `xsl:import-schema` instruction is used within an XSL stylesheet, with default value false)

`http://apache.org/xalan/xslevaluate` (used to enable XSL stylesheet instruction `xsl:evaluate`, with default value false)

These new XSL transformation properties can be set, using Xalan's class `TransformerImpl` when XSL transformation is invoked via API, or via Xalan command line.

1.2) XPath 3.1

XPath version 3.1 specification : <https://www.w3.org/TR/xpath-31/>

- 1) Range "to" expression
- 2) Value comparison operators `eq`, `ne`, `lt`, `le`, `gt`, `ge`
- 3) Function item "inline function expression"
- 4) Dynamic function calls
- 5) "if" expression
- 6) "for" expression
- 7) Quantified expressions 'some', 'every'
- 8) "let" expression

- 9) Sequence constructor expression, using comma operator
- 10) String concatenation operator "||"
- 11) Node comparison operators "is", "<<", ">>"
- 12) Simple map operator '!'
- 13) Instance Of expression
- 14) Implementation of various new XML Schema built-in data types for use in XSLT 3.0 stylesheets and XPath 3.1 expressions. Implementation of, XPath constructor function calls (for e.g, xs:string('hello'), xs:date('2005-10-07') etc) for these supported XML Schema data types.

Following XML Schema built-in types are supported (depicted with XML Schema data type and subtype hierarchy as specified by W3C XML Schema data types specification):

```
xs:anyType
  xs:anySimpleType
    xs:anyAtomicType
      xs:anyURI
      xs:base64Binary
      xs:boolean
      xs:decimal
      xs:integer
        xs:long
          xs:int
            xs:short
              xs:byte
            xs:nonNegativeInteger
              xs:positiveInteger
                xs:unsignedLong
                  xs:unsignedInt
                    xs:unsignedShort
                      xs:unsignedByte
                xs:nonPositiveInteger
                  xs:negativeInteger
          xs:double
          xs:float
          xs:date
          xs:dateTime
          xs:time
          xs:duration
            xs:dayTimeDuration
            xs:yearMonthDuration
          xs:gDay
          xs:gMonth
          xs:gMonthDay
```

- xs:gYear
- xs:gYearMonth
- xs:hexBinary
- xs:QName
- xs:string
 - xs:normalizedString
 - xs:token
 - xs:language
 - xs:Name
 - xs:NCName
 - xs:ID
 - xs:IDREF
 - xs:NMTOKEN

In addition to above mentioned XML Schema built-in data types, an XML Schema type `xs:untyped` specified by XPath 3.1 specification has also been implemented.

15) Collation support

Within the context of XSL languages, a collation is a method by which text information is compared and sorted.

As specified by XPath 3.1 F&O spec, implementations of following collations are available:

15.1) The Unicode Codepoint Collation

15.2) The Unicode Collation Algorithm

Support for following collation uri query parameters is available : 'fallback', 'lang', 'strength'

For the collation's query "lang" parameter, all languages as those supported by Java's 'java.util.Locale' class are available within Xalan's XSLT 3.0 implementation (ref, <https://docs.oracle.com/javase/8/docs/api/java/util/Locale.html>).

For the collation's query "strength" parameter, following values are supported : 'primary', 'secondary', 'tertiary', 'identical'.

15.3) The HTML ASCII Case-Insensitive Collation

16) Sequence type expression

17) Map expression

18) Array expression

19) Cast expression

20) Castable expression

- 21) Treat expression
- 22) Named function reference
- 23) Map and array lookup using function call syntax,
Map and array lookup using unary lookup operator “?”
- 24) Arrow operator (=>)
- 25) Node combination operators union, intersect and except

1.3) XPath 3.1 functions

XPath version 3.1 F&O specification : <https://www.w3.org/TR/xpath-functions-31/>

Implementation of XPath built-in default functions namespace : <http://www.w3.org/2005/xpath-functions>

Implementation of XPath built-in math functions namespace : <http://www.w3.org/2005/xpath-functions/math>

Implementation of XPath built-in map functions namespace : <http://www.w3.org/2005/xpath-functions/map>

Implementation of XPath built-in array functions namespace : <http://www.w3.org/2005/xpath-functions/array>

1) Functions on numeric values

fn:abs

fn:round (implementation of an optional second argument, that's used to specify 'precision')

2) Context functions

fn:current-dateTime

fn:current-date

fn:current-time

fn:implicit-timezone

fn:default-collation

3) Functions giving access to external information

fn:doc

fn:doc-available

fn:collection

fn:unparsed-text

fn:unparsed-text-lines

4) Functions on strings

fn:string-join
fn:upper-case
fn:lower-case
fn:codepoints-to-string
fn:string-to-codepoints
fn:compare (with support for collation argument)
fn:codepoint-equal
fn:contains-token (with support for collation argument)
fn:contains (added support for collation argument)
fn:starts-with (added support for collation argument)
fn:ends-with (with support for collation argument)
fn:substring-before (added support for collation argument)
fn:substring-after (added support for collation argument)

5) String functions that use regular expressions

fn:matches
fn:replace
fn:tokenize
fn:analyze-string

6) Functions that compare values in sequences

fn:distinct-values (with support for collation argument)
fn:index-of (with support for collation argument)
fn:deep-equal (with support for collation argument)

7) Maths trigonometric and exponential functions

math:pi
math:exp
math:exp10
math:log
math:log10
math:pow
math:sqrt
math:sin
math:cos
math:tan
math:asin
math:acos
math:atan
math:atan2

8) Component extraction functions on durations

fn:years-from-duration
fn:months-from-duration
fn:days-from-duration
fn:hours-from-duration
fn:minutes-from-duration
fn:seconds-from-duration

9) Constructing xs:dateTime value

fn:dateTime

10) Component extraction functions on dates and times

fn:year-from-dateTime
fn:month-from-dateTime
fn:day-from-dateTime
fn:hours-from-dateTime
fn:minutes-from-dateTime
fn:seconds-from-dateTime
fn:timezone-from-dateTime
fn:year-from-date
fn:month-from-date
fn:day-from-date
fn:timezone-from-date
fn:hours-from-time
fn:minutes-from-time
fn:seconds-from-time
fn:timezone-from-time

11) Built-in higher-order functions

fn:for-each
fn:filter
fn:fold-left
fn:fold-right
fn:for-each-pair
fn:sort (with support for collation argument)
fn:apply

Dynamic loading and execution, of XSLT stylesheets:

fn:transform

12) Functions on sequences

12.1 General functions on sequences

fn:empty
fn:exists
fn:head
fn:tail

fn:insert-before
fn:remove
fn:reverse
fn:subsequence
fn:unordered

12.2 Aggregate functions

fn:avg
fn:max
fn:min

13) Parsing and serializing

fn:parse-xml
fn:parse-xml-fragment

14) Accessors

fn:node-name
fn:string
fn:data
fn:base-uri
fn:document-uri

15) Functions related to QNames

fn:resolve-QName
fn:QName

16) Functions related to maps

map:merge
map:size
map:keys
map:contains
map:get
map:find
map:put
map:entry
map:remove
map:for-each

17) Functions related to arrays

array:size
array:get
array:put
array:append
array:subarray

array:remove
array:insert-before
array:head
array:tail
array:reverse
array:join
array:for-each
array:filter
array:fold-left
array:fold-right
array:for-each-pair
array:sort (with support for collation argument)
array:flatten

18) Functions on JSON data

fn:parse-json
fn:json-doc
fn:json-to-xml
fn:xml-to-json

Other than the above mentioned newly implemented XPath 3.1 functions, all the functions that are specified for XPath 1.0 are available with Xalan's XPath 3.1 implementation as well.

Please refer to the link <https://www.w3.org/TR/1999/REC-xpath-19991116/> (section "4 Core Function Library") for the details about XPath 1.0 functions.

(2) Features yet not implemented, or partially implemented

XSLT 3.0 instructions

Not yet implemented:

- a) xsl:package
- b) "Conditional Content Construction" instructions xsl:where-populated, xsl:on-empty, xsl:on-non-empty. **XSLT 3.0 spec, ref** : These facilities are available whether or not streaming is in use, but they are introduced to the language specifically to make streaming easier.
- c) XSLT 3.0 map instructions

XSLT 3.0 spec, ref (section 21) : When XSLT 3.0 is used with XPath 3.0, it extends the type system and data model of XPath 3.0 with an additional datatype: the map. The extensions to XPath 3.0 defined within XSLT 3.0 specification section 21 have been incorporated into XPath 3.1. Therefore, when an XSLT 3.0 processor implements the XPath 3.1 Feature, the relevant parts of XSLT 3.0 spec section 21 can be ignored. **Xalan shall not implement this XSLT 3.0 language feature, because Xalan implements XPath 3.1 maps.**

XPath 3.1 features

Partial implementation:

- a) Few XML Schema data types, for use with XPath 3.1 constructor function calls and sequence types are yet not supported

The list of supported XML Schema data types with Xalan, are described within this document's section 1.2) XPath 3.1 point 14) above.

(3) Xalan's XSLT 3.0 and XPath 3.1 test suite details

Xalan XSLT 3.0 development code's latest conformance results with W3C XSLT 3.0 test suite is available here : https://github.com/apache/xalan-java/blob/xalan-j_xslt3.0_mvn/src/test/java/org/apache/xalan/tests/w3c/xslt3/result/w3c_xslt3_testsuite_xalan-j_result.xml

Xalan XSLT 3.0 development code's own XSL test suite is available here : https://github.com/apache/xalan-java/tree/xalan-j_xslt3.0_mvn/src/test and the latest results of these XSL tests are available at locations, https://xalan.apache.org/xalan-j/xsl3/tests/xalan-j_xsl3_test_suite_result.xml & https://xalan.apache.org/xalan-j/xsl3/tests/xalan-j_xsl3_test_suite_result.html

Apache Xalan-J site
<https://xalan.apache.org/xalan-j/>